

MTL106: Probability and Stochastic Processes

(Lecture 15)

February 4, 2026

Example 1: Let X_1, X_2, X_3 be independent and identically distributed (iid) random variables with common PDF

$$f(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Also, let

$$Y_1 = X_1 + X_2 + X_3, \quad Y_2 = \frac{X_1 + X_2}{X_1 + X_2 + X_3}, \quad Y_3 = \frac{X_1}{X_1 + X_2}.$$

Then

$$x_1 = y_1 y_2 y_3, \quad x_2 = y_1 y_2 (1 - y_3), \quad x_3 = y_1 (1 - y_2).$$

The Jacobian of the transformation is given by

$$J = \begin{vmatrix} y_2 y_3 & y_1 y_3 & y_1 y_2 \\ y_2 (1 - y_3) & y_1 (1 - y_3) & -y_1 y_2 \\ 1 - y_2 & -y_1 & 0 \end{vmatrix} = -y_1^2 y_2.$$

Note that

$$0 < y_1 < \infty, \quad 0 < y_2 < 1, \quad 0 < y_3 < 1.$$

Thus, the joint PDF of (Y_1, Y_2, Y_3) is

$$w(y_1, y_2, y_3) = y_1^2 y_2 e^{-y_1} = (2y_2) \left(\frac{1}{2} y_1^2 e^{-y_1} \right), \quad 0 < y_1 < \infty, \quad 0 < y_2, y_3 < 1.$$

It follows that Y_1, Y_2 , and Y_3 are independent.

Moment Generating Function of $(X_1, X_2, \dots, X_n)$: Let $X_1, X_2, \dots, X_n$ be jointly distributed. If

$$E \left[\exp \left\{ \sum_{j=1}^n t_j X_j \right\} \right]$$

exists for $|t_j| \leq h_j$, $j = 1, 2, \dots, n$, for some $h_j > 0$, we write

$$M(t_1, t_2, \dots, t_n) = E[e^{t_1 X_1 + t_2 X_2 + \dots + t_n X_n}] \quad (1)$$

and call it the MGF of the joint distribution of $(X_1, X_2, \dots, X_n)$, or simply, the MGF of $(X_1, X_2, \dots, X_n)$.

Characteristic Function of $(X_1, X_2, \dots, X_n)$: Let $t_1, t_2, \dots, t_n$ be real numbers and $i = \sqrt{-1}$. Then the characteristic function of $(X_1, X_2, \dots, X_n)$ is defined by

$$\phi(t_1, t_2, \dots, t_n) = E \left[\exp \left(i \sum_{j=1}^n t_j X_j \right) \right]. \quad (2)$$

As in the univariate case, $\phi(t_1, t_2, \dots, t_n)$ always exists.

Theorem 1. The MGF $M(t_1, t_2)$ uniquely determines the joint distribution of (X, Y) and, conversely, if the MGF exists, it is unique.

Remark 1. The MGF $M(t_1, t_2)$ completely determines the marginal distributions of X and Y . Indeed,

$$M(t_1, 0) = E[e^{t_1 X}] = M_X(t_1), \quad (3)$$

$$M(0, t_2) = E[e^{t_2 Y}] = M_Y(t_2). \quad (4)$$

Theorem 2. If the MGF $M(t_1, t_2)$ exists, the moments of all orders of (X, Y) exist and may be obtained from

$$\left. \frac{\partial^{m+n} M(t_1, t_2)}{\partial t_1^m \partial t_2^n} \right|_{t_1=t_2=0} = E(X^m Y^n). \quad (5)$$

Thus,

$$\begin{aligned} \frac{\partial M(0, 0)}{\partial t_1} &= E(X), & \frac{\partial M(0, 0)}{\partial t_2} &= E(Y), \\ \frac{\partial^2 M(0, 0)}{\partial t_1^2} &= E(X^2), & \frac{\partial^2 M(0, 0)}{\partial t_2^2} &= E(Y^2), \\ \frac{\partial^2 M(0, 0)}{\partial t_1 \partial t_2} &= E(XY), \end{aligned}$$

and so on.

Theorem 3. Two random variables X and Y are independent if and only if

$$M(t_1, t_2) = M(t_1, 0)M(0, t_2), \quad \forall t_1, t_2 \in \mathbb{R}. \quad (6)$$

Example 2: Let X be a random variable with PDF

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty.$$

Let $Y = X^2$. Then

$$M_Y(t) = E[e^{tX^2}] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(1/2)(1-2t)x^2} dx = \frac{1}{\sqrt{1-2t}}, \quad t < \frac{1}{2}.$$

Theorem 4. Let X_i , $i = 1, 2, \dots, n$, be independent random variables with respective MGFs $M_i(t)$. Then the MGF of

$$Y = \sum_{i=1}^n a_i X_i$$

for real numbers $a_1, a_2, \dots, a_n$ is

$$M_Y(t) = \prod_{i=1}^n M_i(a_i t).$$

Remark 2. If X_i are iid, then the MGF of

$$Y = \sum_{i=1}^n X_i$$

is

$$M_Y(t) = [M(t)]^n.$$

Example 3: Let X_i , $i = 1, 2, \dots, n$, be iid random variables with common PMF

$$P\{X = k\} = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \dots, n, \quad 0 < p < 1.$$

Then

$$M(t) = E[e^{tX}] = \sum_{k=0}^n e^{tk} \binom{n}{k} p^k (1-p)^{n-k} = (1-p+pe^t)^n.$$

It follows that the MGF of

$$S_m = X_1 + X_2 + \dots + X_m$$

is

$$M_{S_m}(t) = (1-p+pe^t)^{nm}.$$

Covariance: If $E[(X - E(X))(Y - E(Y))]$ exists, we call it the covariance between X and Y and write

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))] = E(XY) - E(X)E(Y).$$

Note that if X and Y increase (or decrease) together, the covariance is positive, whereas if one increases while the other decreases, the covariance is negative. Thus, $\text{Cov}(X, Y)$ provides a measure of association or joint variation between X and Y .